

describes and defines how network services are designed, constructed, and deployed using virtualized software components and how these are decoupled from the hardware upon which they execute.”

In Figure 1, we can see how traditional network devices such as routers, VPNs, load balancers, and others can be first converted to software-based functions and then placed on standard switches and servers. These functions run as virtual applications over the decoupled hardware. This NFV-based approach does not require specialised hardware.

NFV and SDN

SDN (software defined network) is a component and enabler of NFV. It paves the way for virtualization and orchestration. With the advent of SDN came the separation of the control and data plane, which served NFV by providing programmable connectivity between VNFs. These connections are further managed by the orchestrator of the VNFs, which acts as an SDN controller. NFV can be implemented without SDN too, but the two are combined for greater performance and scalability, and to facilitate operations and maintenance procedures.

Advantages of NFV

NFV brings a lot to the table. If implemented widely, it can enable great flexibility and scalability in telecom infrastructure.

Scalability: By leveraging the power of NFV, we can make a serious dent in capex and opex expenditures. The operator need not look into network scalability demands but only concentrate on consumers and cater to their service usage.

Flexibility: With software-based applications, it is easy to seamlessly integrate new services without any prerequisites. NFV assures higher adaptability and easy installation.

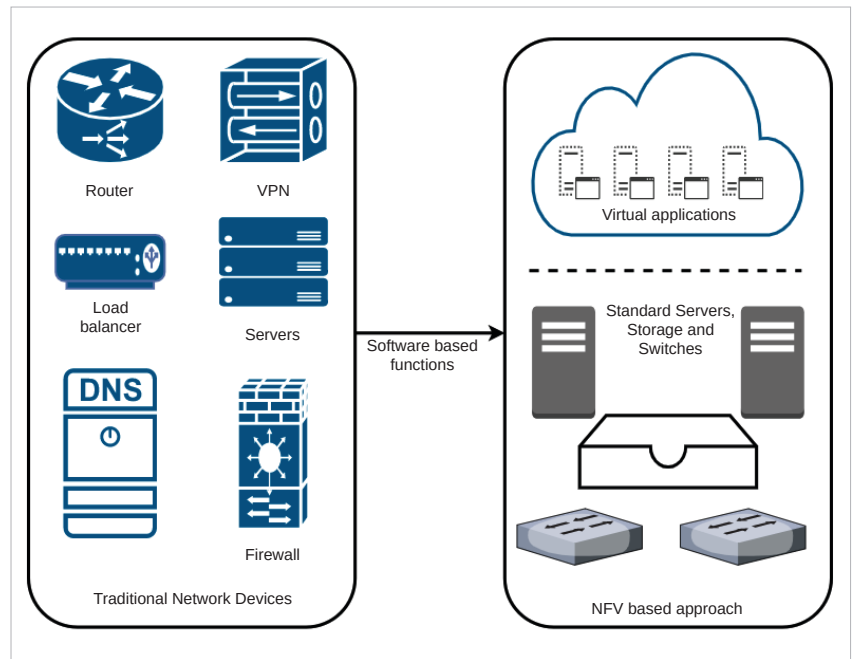


Figure 1: NFV-based approach compared to the traditional approach in telecom infrastructure

Time: The installation time too can be narrowed down with automation and tools.

Cost: The expense of buying new network devices, their cooling equipment, the space to keep them, and the power to operate them comes down steeply with NFV. Much of the software is accessible under an open source licence.

Security: With NFV, operators are able to run software-based virtual firewalls between the networks. They have greater control over the traffic and services between the networks.

NFV infrastructure management

OpenStack, the popular open source cloud computing software project, is a good solution for NFV infrastructure management. It provides the APIs to service computing, storage, and networking. This IaaS platform provides an interface to control the NFV infrastructure, which is deployed between the virtual machines and other networking

hardware such as switches, routers, and load balancers. The life cycle of virtual machines is also carried out on this platform — they can be deployed, managed, scaled, and destroyed on it.

NFV infrastructure is all about virtualization and OpenStack does provide that with its various components, as shown in Figure 2. Many telecom companies recognise the potential of OpenStack as it offers a choice to manage and orchestrate all these virtual functions. Ninety-six per cent of cloud service providers (CSPs) implementing NFV strategies say OpenStack is essential to their success (<https://www.openstack.org/use-cases/telecoms-and-nfv/>). The open, modular and interoperable framework of the OpenStack project offers telecoms and enterprises the ability to design the NFV system of their choosing, without using unnecessary components.

But that’s not all when it comes to NFV technology. VMs are fine but the technology is a